

# SOUFYANE ELAOUNI

Mechatronics Engineering Student | Automotive Embedded Systems

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Tetouan, Morocco  
📍 National mobility



## PROFILE

Final-year Mechatronics Engineering student with a strong focus on **automotive engineering**. Backed by industrial experience at **Renault Group** in developing diagnostic tools (**CAN/UDS**) and testing Electronic Control Units (ECUs), I combine skills in hardware electronics, embedded programming, and system simulation. Seeking a 6-month graduation internship (PFE) to advance in embedded systems, automotive validation (**HIL/SIL**), or systems engineering.

## EDUCATION

State Engineering Degree in Mechatronics – ENSA Tetouan

2023 – Present

Preparatory Classes for Engineering Schools (Mathematics & Physics)

2021 – 2023

High School Diploma in Mathematics

2021

## PROFESSIONAL EXPERIENCE

Embedded Systems & Automotive Diagnostics Intern

Jun 2026 – Sept 2026

Renault Technology Morocco (RTMA)

Tangier, Morocco

- Replaced a manual ECU inspection bottleneck (VCI/DDT2000) by designing a standalone ESP32 test bench and custom "Plug & Play" wiring harnesses based on vehicle CAN topologies.
- Developed the C/C++ firmware, implementing the ISO-TP (ISO 15765-2) transport layer and UDS (ISO 14229) protocol to interrogate multiplexed networks (BCM, Gateway, etc.).
- Built an autonomous IoT architecture (embedded Wi-Fi web server, LittleFS for CSV traceability) with a real-time dashboard, reducing ECU batch validation time from **10 minutes to 19 seconds**.

Systems Engineering & Industrial Automation Intern

Jul 2025

Holcim Morocco

Oujda, Morocco

- Reverse-engineered an obsolete electronic sequencer to automate a critical dust filtration system, designing new control logics (Standard,  $\Delta P$ , and Maintenance) to prevent material loss.
- Programmed a Siemens S7-1200 PLC (LADDER / TIA Portal) to interface with industrial pressure and temperature sensors, driving existing electrovalves for bag unclogging while preventing overheating.
- Developed a local HMI (WinCC) and integrated real-time data into the plant's global SCADA system via OPC, successfully validating the architecture on a 12-electrovalve prototype filter.

## TECHNICAL SKILLS

**Embedded** : ESP32, STM32, Raspberry Pi, Altium Designer (PCB)

**Programming** : C/C++, Python

**Automotive** : CAN, CAN FD, UDS, ISO-TP, OBD-II

**Tools** : Git, Linux, VS Code

**Simulation** : MATLAB/Simulink, Stateflow, PID, SMC, EKF

**Industrial** : TIA Portal, WinCC, SCADA, OPC

## ENGINEERING PROJECTS

Battery Management System (BMS) - Model-Based Design

MATLAB/Simulink • Stateflow • EKF Estimation

- Modeled 400V Li-ion electro-thermal dynamics and implemented real-time SoC estimation (Extended Kalman Filter & Coulomb Counting).
- Designed Stateflow safety logics (current limiting, fault protection) to prevent overvoltage, deep discharge, and thermal runaway.

Adaptive Cruise Control (ACC) - Model-Based Design

MATLAB/Simulink • AUTOSAR • ISO 26262

- Developed a hybrid PID controller for vehicle longitudinal dynamics (speed/spacing regulation) using simulated 77GHz RADAR data.
- Architected AUTOSAR-compliant SWCs with ISO 26262 (ASIL C) fail-safe modes, validated via Software-in-the-Loop (SIL) testing.

Smart In-Vehicle Infotainment (IVI) & Body Control System

ESP32 • STM32 • CAN Bus • IoT Dashboard

- Architected a distributed automotive network linking an STM32 (managing 7+ peripherals: NFC ignition, optical speed, rain sensors, HVAC actuation) and an ESP32 (cockpit UI & IoT cloud telemetry) via CAN bus.

## LEADERSHIP & ACTIVITIES

- Training Manager – Mechatronics Club** : Led embedded systems & IoT training (Arduino, ESP32, C++) for over 60 engineering students, culminating in an HTTP-based smart weather station. Coordinated with 3 industry experts to deliver masterclasses on IIoT and cybersecurity.
- Technical Lead – National Robotics Competition (MechaWorldCup)** : Authored technical specifications and shortlisted the Top 10 teams. Directed an intensive technical/pitch bootcamp and moderated the final event and jury evaluations.

## LANGUAGES & CERTIFICATIONS

### Languages

**English** : Fluent  
**French** : Fluent  
**Arabic** : Native

### Certifications

- ISO 26262 (Functional Safety) – Ansys (2026)
- Electrical Engineering Job Simulation – GE Aerospace (2026)
- Industrial Robotics ABB – Udemy (2025)